

DRONE



**TEACHER:
CERTIFIED
TRAINER**

**DURATION:
12 CLASSES (1 HOUR PER
CLASS)**

**MODE:
OFFLINE**

This is a powered, mechanical and hands-on learning solution focused on drones, where experiments connect textbook concepts to real-world applications. Each session involves building, assembling and testing drone components and complete flying models, both motorised and electronic, to understand how drones work in real-life scenarios. Students learn core STEM concepts such as aerodynamics, propulsion, electronics, sensors and flight control systems through practical activities. High-quality imported drone kits, motors, frames and electronic components made from durable, industry-grade materials are used to ensure safe, reliable and professional learning experiences.

COURSE CURRICULUM

1. Discover the Drone World

- Introduction to drones and their everyday uses (delivery, photography, rescue, agriculture).
- Understanding drones as flying machines.

2. Anatomy of a Drone

- Frame: Structure that holds all drone components together.
- Motors & Propellers: Components responsible for lift and movement.
- Battery: Power source of the drone.
- Controller & Receiver: Devices used to control the drone wirelessly.

3. Secrets of Flight

- Basics of lift, thrust, drag and gravity.
- How propellers push air to make the drone fly.

4. Let's Build – Part 1

- Identifying drone parts and tools.
- Assembling the drone frame and motor placement.

5. Let's Build – Part 2

- Attaching propellers and battery safely.
- Understanding balance and symmetry in drone building.

6. Mastering the Controls

- Learning basic drone movements: up, down, left, right, forward and backward.
- Introduction to joystick controls and directions.

7. Ignition and Inspection

- Pre-flight safety checks.
- Battery check, propeller inspection and safe handling practices.

8. Into the Skies

- First guided flight experience.
- Understanding take-off, hovering, and landing.

9. Pilot in Command

- Learning responsibility and rules of safe flying.
- Developing focus, coordination, and confidence while flying.

10. Aerial Fun Missions – Part 1

- Simple flying challenges like hovering at a spot.

- Path-following and controlled movement activities.

11. Aerial Fun Missions – Part 2

- Obstacle avoidance and target landing missions.
- Team-based flying activities to improve coordination.

12. Drone Show and Awards

- Demonstration of learned skills through a mini drone show.
- Recognition and awards for participation and achievements..

Class Structure

1. Introduction to Drone Components

- Learn about basic drone parts such as motors, propellers, batteries, and controllers.
- Understand their function in making a drone fly.

2. Supervised Drone Assembly

- Build a beginner-friendly drone using high-quality imported kits.
- Assembly done under guidance to ensure safety and correctness.

3. Understanding Flight Concepts

- Simple explanation of aerodynamics and motion.
- Connecting drone flight concepts to real-world flying objects like birds and airplanes.

4. Practical Flying Activities and Challenges

- Fun challenges including:
- Take-off and landing tasks
- Hover control games
- Direction control missions
- Hands-on flying practice to build confidence

5. Safety, Discipline and Real-Life Applications

- Learning drone safety rules and responsible usage.
- Understanding how drones are used in real-life professions.